

Kavya Chandrika Vempalli

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PROFILE SUMMARY

Software Engineer with **4+ years of experience** in designing, developing, and deploying scalable full-stack and backend systems. Strong expertise in **Java, Spring Boot, React, TypeScript, AWS, and microservices architectures**. Proven ability to deliver high-performance applications, optimize system reliability, and collaborate effectively in Agile environments. Experienced in cloud deployments, containerization, and building secure, production-ready solutions across enterprise and consumer platforms.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, C#, **Java**, **Python**, Kotlin, JavaScript, TypeScript, HTML, CSS, YAML
- **Frameworks & Databases:** **Spring Boot**, Spring MVC, React.js, Node.js, SQL, PostgreSQL, JSON
- **Cloud & DevOps:** AWS, Docker, Kubernetes, Jenkins, CI/CD
- **Tools & Practices:** Linux, Git, GitHub, GitLab, Bitbucket, **Agile**, REST APIs, JUnit, Mockito, Maven, JIRA, Confluence
- **Big Data & Streaming:** Hadoop, Spark
- **Python Libraries:** NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, OpenCV, Flask, FastAPI, Django, Pytest

CERTIFICATIONS

- AWS Certified Developer – **Associate**.

PROFESSIONAL EXPERIENCE

EyCrowd Inc | Software Engineer Intern (Remote)

July 2025 – Aug 2025

- Performed end-to-end application testing, identifying and resolving critical bugs to improve system stability and performance.
- Integrated **social media APIs** in React Native to enable secure account linking and social authentication.
- Designed and implemented **friend management workflows** including requests, approvals, and real-time list updates.
- Built native **photo and video capture** features using device camera APIs while optimizing storage and performance.
- Collaborated with cross-functional teams to deliver features aligned with product requirements and timelines.

Arizona State University | Graduate Software Engineer, Tempe, AZ

Aug 2024 – May 2025

- Developed reusable **React + TypeScript UI components**, reducing content creation effort by over **50%**.
- Reviewed 30+ GitHub pull requests and resolved 10+ high-severity production issues, significantly improving application reliability.
- Deployed services on **AWS (EC2, S3, Cognito)** and containerized applications using **Docker** for scalable deployments.
- Built HR modules including **leave management, attendance tracking, and payroll integrations** using Spring Boot and SQL.
- Implemented overtime and workflow automation systems, reducing approval turnaround time and manual processing errors.

Hyundai Mobis | Research Engineer - Hyderabad, India

Feb 2021 – Jul 2023

- Designed and delivered **scalable Spring Boot microservices** for In-Vehicle Infotainment (IVI) systems supporting real-time navigation and media streaming.
- Built low-latency **real-time chat systems** using WebSockets and Redis, achieving sub-200 ms message delivery.
- Implemented secure file transfer and volatile messaging with TTL policies to meet compliance and data retention standards.
- Containerized backend services using **Docker**, improving deployment consistency and reducing environment-related issues by 30%.
- Led CI/CD and configuration management using Git pipelines, ensuring reliable releases across development cycles.
- Automated system monitoring and debugging with Python scripts, cutting manual intervention by **40%**.

Quanint Tech Soft Pvt Ltd | Web Development Intern - Hyderabad, India

Jun 2019 – Jul 2019

- Developed responsive web applications using **HTML, CSS, JavaScript, Bootstrap, and React.js**, increasing user engagement by 25%.
- Optimized website performance, improving page load times by 30%.
- Delivered WordPress-based front-end solutions tailored to client requirements, enhancing overall user experience.

PROJECTS

Expense Tracker (Full-Stack Web Application)

May 2025

- Developed a full-stack expense management system using **React, TypeScript, and Spring Boot** with complete CRUD functionality.

- Implemented advanced filtering, search, and sorting to deliver real-time analytics with zero redundant API calls.

Tech Stack: React, TypeScript, Spring Boot, REST APIs.

Document Redaction Office Add-in

Dec 2025

- Built a Microsoft Word Add-in using **React 18 and TypeScript** to detect and redact 14 types of sensitive information.
- Implemented tier-based pattern matching for secure redaction while maintaining document compliance and audit trails.

Tech Stack: React, TypeScript, Vite, Office.js API, Fluent UI.

Real-Time Cryptocurrency Price Streaming Platform

Sep 2025

- Developed a real-time crypto price streaming platform using **Next.js, Node.js, and TypeScript**.

- Implemented server-side streaming and browser automation for sub-second data propagation.

Tech Stack: Next.js, Node.js, TypeScript, Playwright, ConnectRPC.

AI-Powered MCQ Generator

Oct 2025

- Built an automated MCQ generation system using **OpenAI GPT-4.1-nano API** for educational content.

- Designed intelligent text chunking and validation layers to ensure high-quality question generation.

Tech Stack: Python, OpenAI API, PyPDF2, JSON.

Patient Dashboard – Healthcare Analytics Interface

Sep 2025

- Developed a responsive healthcare dashboard to visualize patient vitals and medical records.

- Implemented interactive trend analysis using Chart.js with real-time data filtering.

Tech Stack: HTML, CSS, JavaScript, Chart.js, REST APIs.

Agriculture Pest Classification using CNNs

May 2024

- Built a CNN model using **ConvNeXt**, achieving **95% classification accuracy** across 12 pest classes.

- Performed dataset curation and augmentation to improve model robustness.

Tech Stack: PyTorch, Python, NumPy, Pandas.

Face Recognition using AWS

May 2024

- Designed and deployed cloud architecture using **AWS EC2, S3, Lambda, and SQS** for face detection.

- Implemented video-based face detection workflows with scalable processing.

Tech Stack: AWS, Flask, Python.

Westside Website Revamp (HCI Project)

Dec 2023

- Conducted usability testing and heuristic evaluations to improve task completion rates.

- Analyzed post-task metrics to measure UI improvements.

Tech Stack: Axure, UX Research, MATLAB.

Guardian Angel (Mobile Computing Project)

Dec 2023

- Developed a real-time safety advisory system using physiological and vehicular data.

- Integrated IoT data streams via ThingSpeak with mobile application processing.

Tech Stack: MATLAB, Simulink, Java, Android.

EDUCATION

Master of Science in Computer Science | Arizona State University — Tempe, AZ

Aug 2023 – May 2025

GPA: 3.77 / 4.00

Bachelor of Science in Computer Science and Engineering

Aug 2016 – Oct 2020

Osmania University — Hyderabad, India

GPA: 8.13 / 10.00